EXHAUSTING A FINITE SIMPLICIAL SET BY GLUING SINGLE SIMPLICES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: W is visibly a simplicial set (the set of degeneracies of a fixed simplex is stable under all simplicial operators); the dependence on Lemma lemma-glue-simplex is a navigation stub to the registered card.

Editorial note. The label lemma-add-simplices names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Gluing a single simplex. Same unit, Lemma lemma-glue-simplex. Each step of the chain U = W^0 subset W^1 subset ... subset W^r = V must satisfy the hypotheses of Lemma lemma-glue-simplex, so that each inclusion is a pushout of a boundary inclusion.

Source reference. Original reference 018R: sources/stacks-simplicial/pinned-original-source.tex, source lines 3301–3301. Prerequisite pointer invoked at proof line(s) 3354; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $U \subset V$ be simplicial sets, with U_n, V_n finite nonempty for all n. Assume that U and V have finitely many nondegenerate simplices. Then there exists a sequence of sub simplicial sets*

$$U = W^0 \subset W^1 \subset W^2 \subset \ldots W^r = V$$

such that Lemma 018R applies to each of the inclusions $W^i \subset W^{i+1}$.

Proof. Let n be the smallest integer such that V has a nondegenerate simplex that does not belong to U. Let $x \in V_n$, $x \notin U_n$ be such a nondegenerate simplex. Let $W \subset V$ be the set of elements which are either in U, or are a (repeated) degeneracy of x (in other words, are of the form $V(\varphi)(x)$ with $\varphi : [m] \to [n]$ surjective). It is easy to see that W is a simplicial set. The inclusion $U \subset W$ satisfies the conditions of Lemma 018R. Moreover the number of nondegenerate simplices of V which are

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not contained in W is exactly one less than the number of nondegenerate simplices of V which are not contained in U. Hence we win by induction on this number. $\square$